

CryptoCopilot

Signal from the Noise: A Polyglot Intelligence
Engine for the Crypto Market.

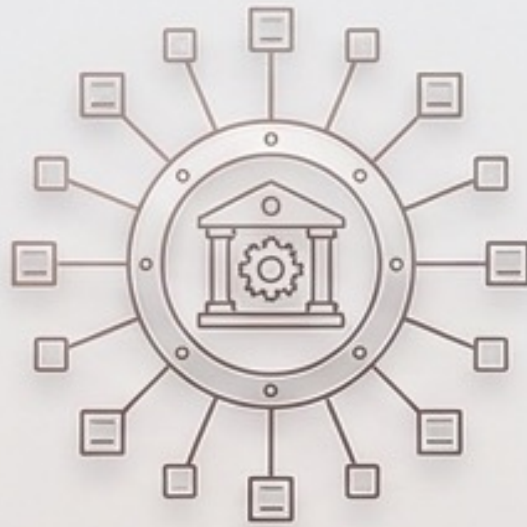
Built by Kamran Zajkani

[ML Signals]

[Technical Analysis]

[Cited AI Chat]

[Safe Paper Trading]



Traditional Finance (TradFi)



Controlled by central banks.



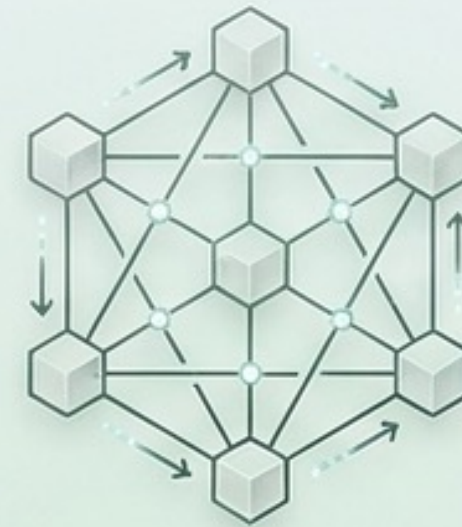
High fees to hold and move your own money.



Custodial: Someone else is always in charge of your account.

Centralized Control

The Mission: Turn scattered market data and chaotic news into clear information and deterministic signals.



Cryptocurrency



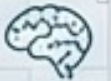
Decentralized, open, and global.



Self-custody: You hold your own money directly.

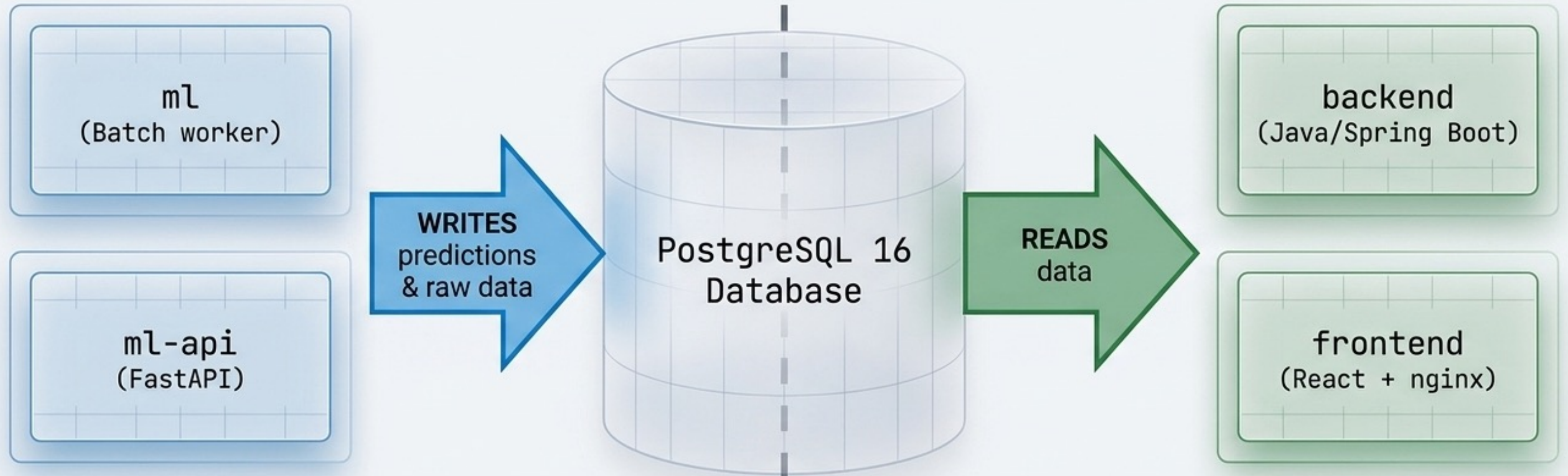


Volatile, complex, and overwhelmingly noisy for newcomers.



**True Freedom, but
High Complexity**

One System, Two Languages

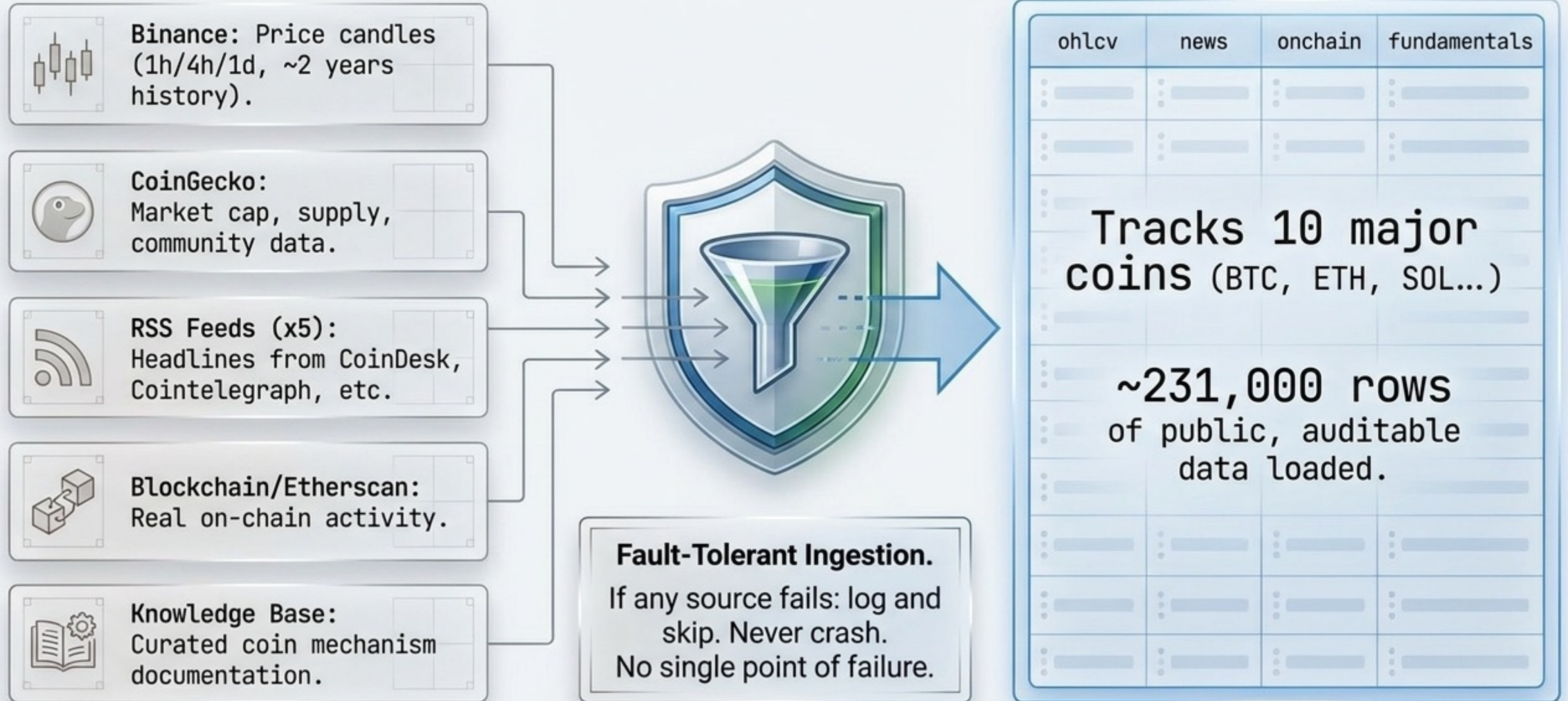


Ingests data, runs XGBoost Machine Learning, calculates SHAP drivers.

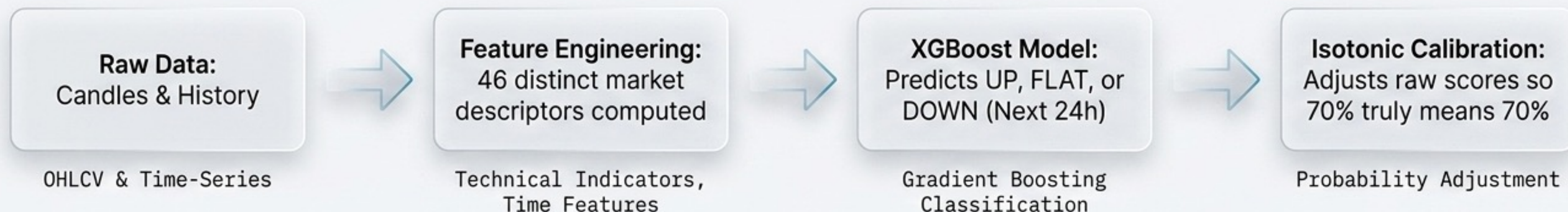
Real-time API, Technical Analysis, RAG AI Chat, UI serving.

THE POLYGLOT BOUNDARY: NO DIRECT CALLS.
Python only writes; Java only reads.

Robust & Auditable Data Ingestion



The Assembly Line



The 'X-Ray' Explanation

An Honest Edge, Not a Crystal Ball. ROC-AUC = 0.578.
A small but real statistical edge above pure luck.

Calibrated Probabilities.

Explainable AI. SHAP reveals the exact top-3 features driving every prediction, proving it is not a black box.



Technical Analysis: A Verifiable Second Opinion

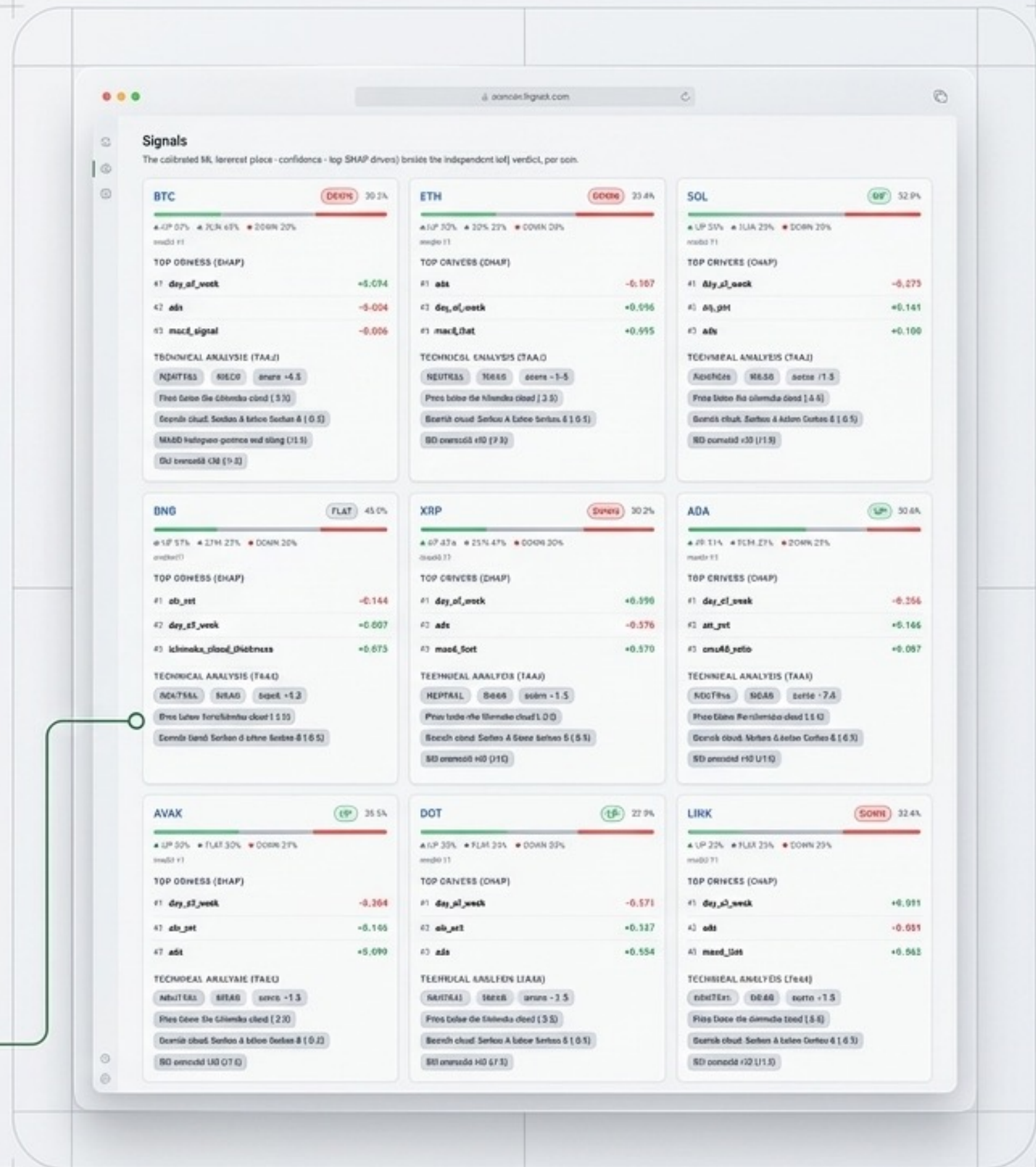
The Input: Raw price candles straight from the database.

The Engine: Computed in Java using ta4j (independent of the Python ML model).

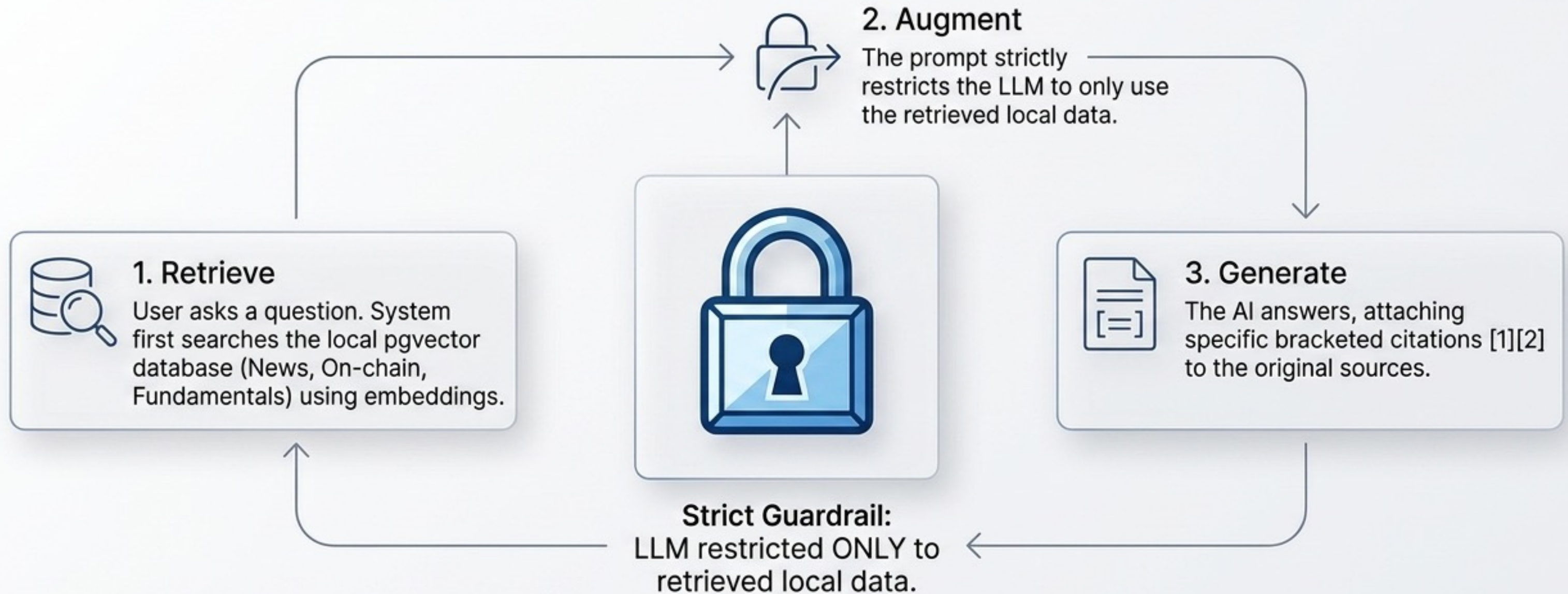
The Indicators: Ichimoku Cloud, RSI, MACD, Bollinger, ATR.

The Output: A deterministic verdict (BULLISH, NEUTRAL, BEARISH) with confidence levels.

100% Transparent.
Shows the exact rules that fired.
You can audit every step.



The Researcher: AI That Actually Cites Its Sources



Performance:

`recall@8 = 0.90` | 100% of answers cited.



Safety:

Refuses if answer is absent.
Never gives trading advice.



Hardware Toggle:

Runs free & local (Ollama llama3.2:3b) or switches to cloud (OpenAI gpt-4o-mini) with one click.

 **ML Signal**
(Probabilities)

 **Technical**
(ta4j indicators)

 **Fundamentals**
(On-chain health)

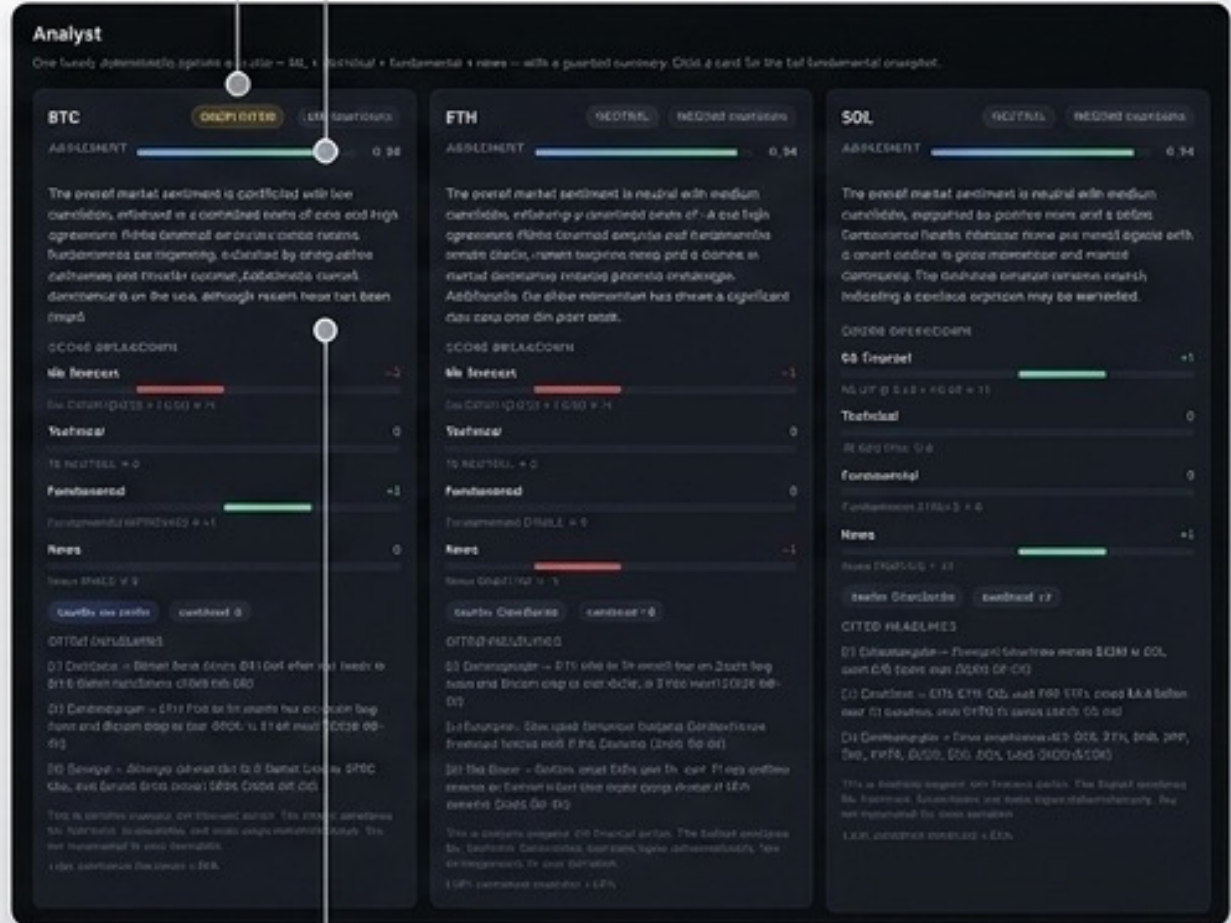
 **News**
(AI sentiment)


The Analyst
A deterministic, rule-based engine. Converts each stream into a rigid score from -2 to +2.

1 Direction & Conviction

2 Agreement Score
(How aligned the four lenses are)

3 Guarded AI Summary
(Safe, truthful synthesis of the numbers)



The Analyst dashboard displays analysis for three cryptocurrencies: BTC, ETH, and SOL. Each section includes an 'ASSESSMENT' bar, a 'Direction & Conviction' score, an 'Agreement Score' (how aligned the four lenses are), and a 'Guarded AI Summary' (safe, truthful synthesis of the numbers). The dashboard also features a 'Scores Breakdown' section with metrics like 'ML Forecast', 'Technical', 'Fundamental', and 'News'.

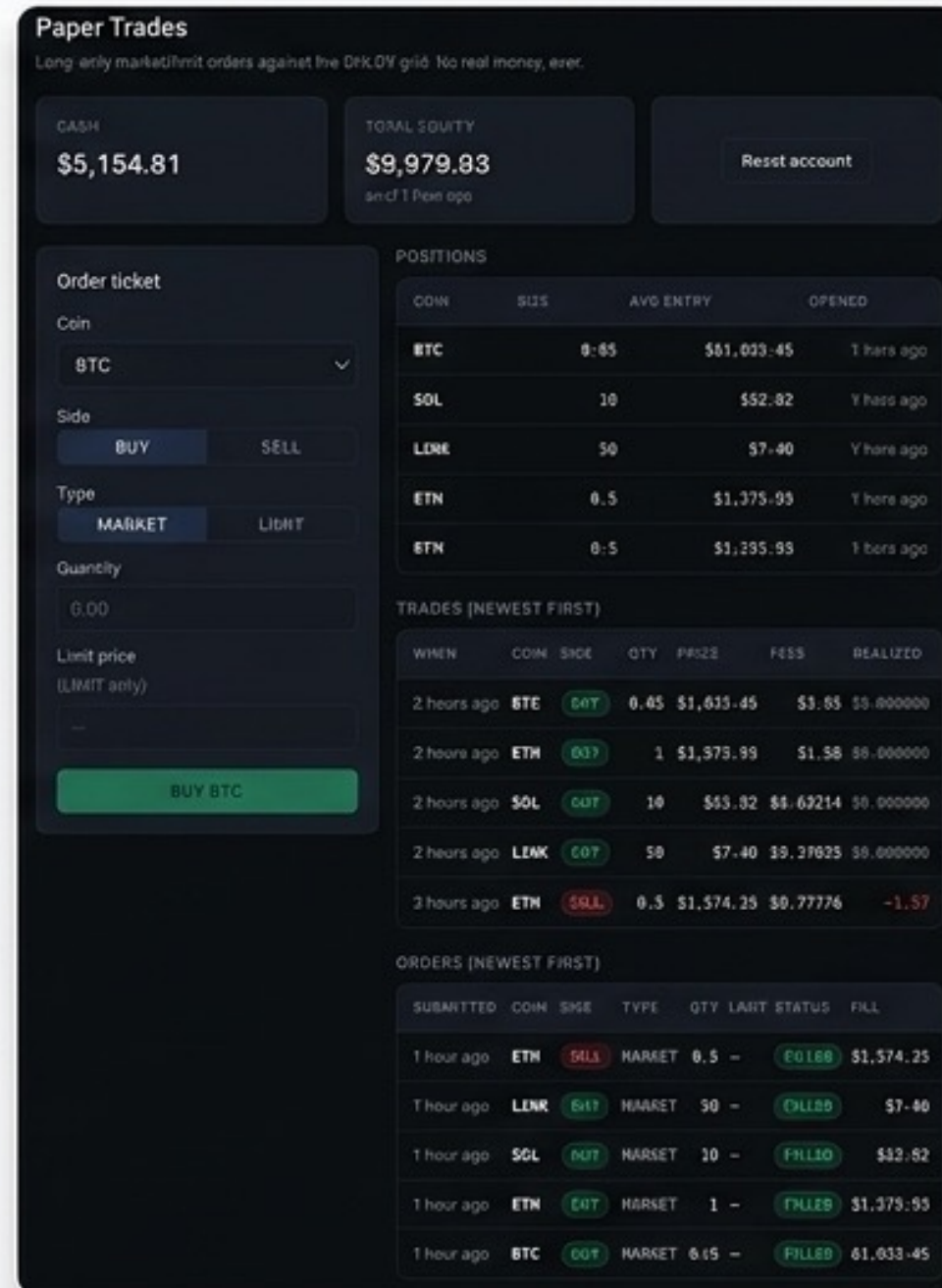
Acting Safely: The Paper Trading Engine

The Mechanics

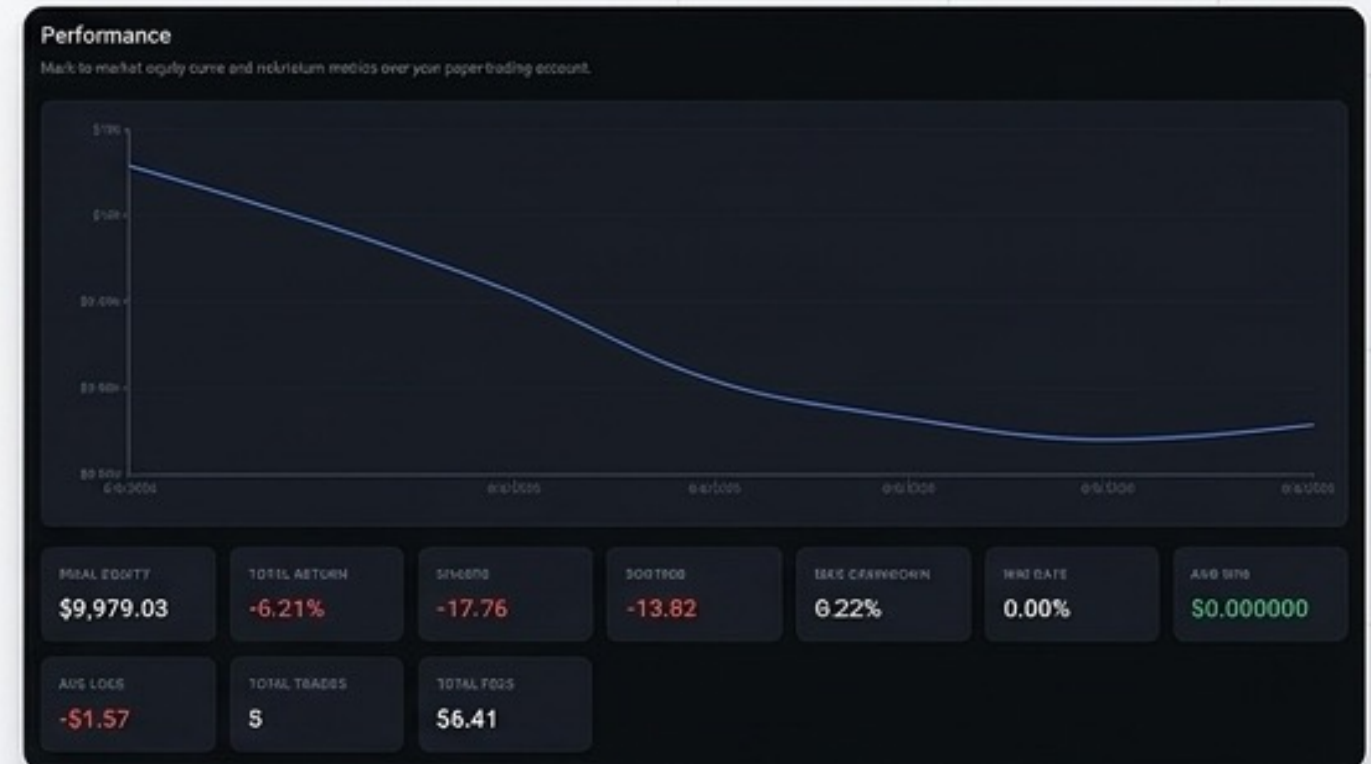
Fake Money, Real Mechanics: Practice execution with zero financial risk.

Honest Fills: Simulates realistic slippage and exacts a 0.1% exchange fee per trade.

Full Tracking: Long-only market and limit orders, position tracking, and detailed P&L.



The Reality Check



! Honest Engineering:
The default test strategy is unprofitable (-0.21% return). The goal was to build a mathematically flawless tracking engine, not a "get rich quick" magic formula.

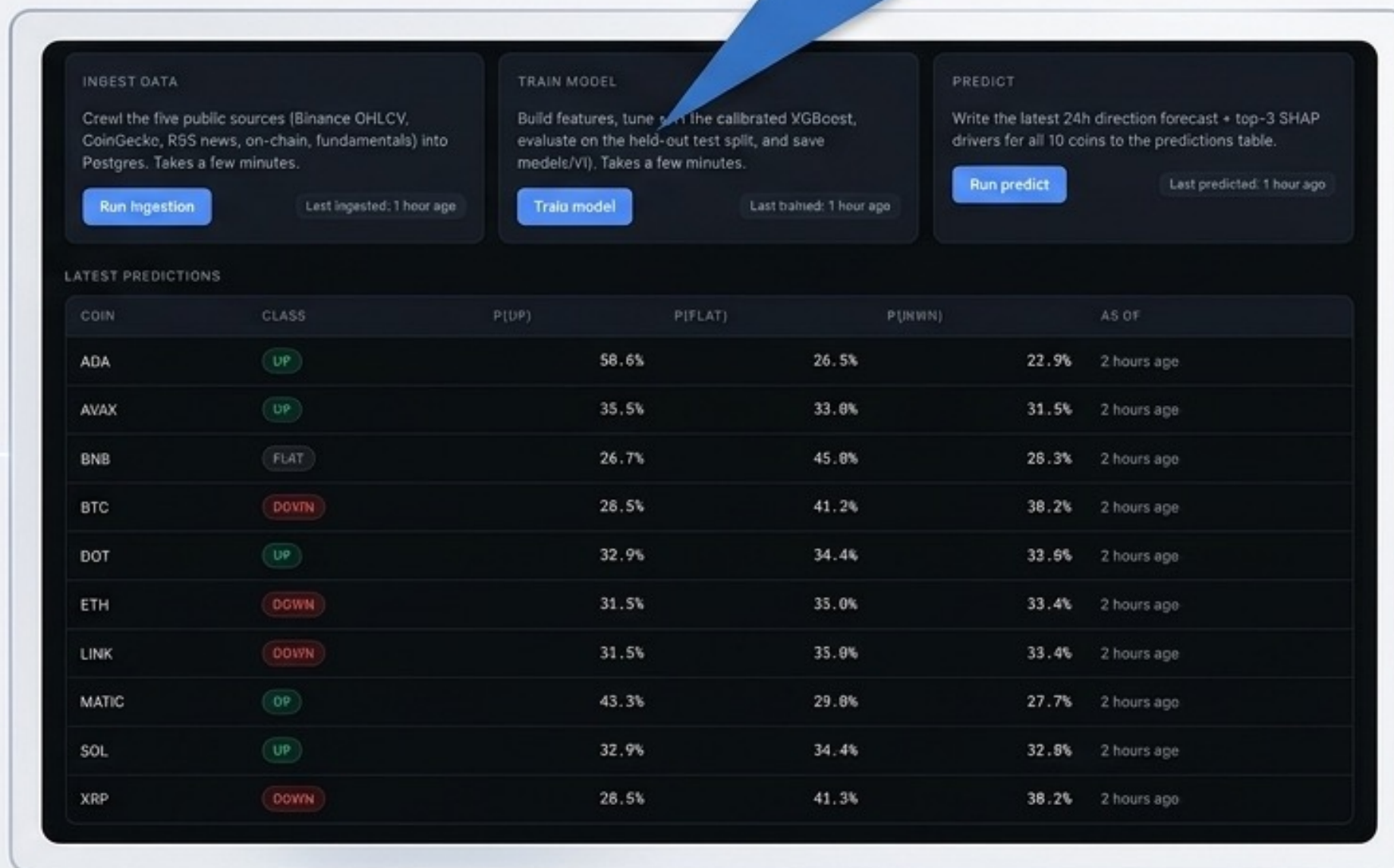
The Application: Clean, Typed, and Browser-Controllable

Run the Pipeline In-Browser:
Trigger Data Ingestion, Model Training, and Prediction jobs on demand without ever touching a command line.

Frontend: React + Vite + TypeScript.
Fast, responsive, strictly typed.

Charting: klinecharts for interactive candlesticks (Ichimoku overlays).

Scope: 7 complete pages
(Markets, Signals, Analyst, Chat, Trades, Performance, Pipeline).



⚠ A persistent decision-support disclaimer is rendered on every single page.

Machine Learning

ROC-AUC 0.578

A small, genuine statistical edge above random luck. F1 score limited by ~2 years of data history. Calibrated, explainable, and un-exaggerated.

AI Retrieval (RAG)

recall@8
0.90 | 100%
Citations

Strong grounded retrieval. Refuses out-of-corpus questions. Strictly enforces the “no trading advice” guardrail. Local inference runs for €0.

Architecture

5 Containers,
3 Languages,
1 DB

Tested, automated, polyglot CI/CD. The paper trading engine is mathematically correct.

The goal was never to beat the market. The goal was to build a production-grade system, end-to-end, and to report the numbers honestly.